

Characters: Epistemic State Modeling, Verse–Prose Alternation, and Sociolect Alignment in Shakespeare’s Complete Dramatic Canon

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Abstract

Conversational roleplay agents driven by Large Language Models (LLMs) present a recurring structural failure when applied to classic dramatic literature: they suffer from omniscient anachronism, prosodic collapse, and the flattening of historical social registers. In this paper, we present CHARACTERS, an epistemic dialogue and computational modeling architecture constructed over the complete 37-play canonical corpus of William Shakespeare (1590–1613). By extracting line-level stage presence from TEI-XML markup via a bipartite character-witness graph, our framework deterministically bounds character retrieval to witnessed diegetic events, thereby preserving dramatic irony across acts and scenes. Furthermore, we integrate a dynamic verse–prose selector and an automated Early Modern English prosodic scansion engine. Using a local Small Language Model (Qwen 2.5-3B), we evaluate the framework against high-risk dramatic irony probes in *Hamlet*, *Othello*, *Twelfth Night*, and *Macbeth*. Our empirical results demonstrate that chronologically bounded retrieval successfully preserves character ignorance while exposing a critical methodological distinction for Digital Humanities: the divergence between semantic ignorance preservation and prompt-induced lexical echoing.

Keywords: Computational Literary Studies, Digital Humanities, Shakespeare, Epistemic Modeling, Dramatic Irony, Prosodic Scansion, Early Modern English

1 Introduction

In dramatic theory, dramatic tension is fundamentally driven by *epistemic asymmetry*—discrepancies between what the audience knows, what individual characters witness, and what remains concealed (Pfister 1991). Classic tragedy and comedy depend upon this bounded perspective: Hamlet’s hesitations are governed by the uncertainty of the Ghost’s testimony; Othello’s descent is predicated on his ignorance of Iago’s domestic manipulations; and Olivia’s affections in *Twelfth Night* rely on her unawareness of Viola’s disguise.

However, modern generative language models (LLMs) treat literary corpora as undifferentiated semantic stores. When prompted to adopt a literary persona, models routinely exhibit omniscient anachronism: characters foresee their ultimate demise or reveal plot secrets long before their diegetic revelation. Furthermore, commercial dialogue engines fail to navigate the formal structural alternation between blank verse (unrhymed iambic pentameter) and comic/colloquial prose, while ignoring historical pronominal sociolects (the Early Modern *thou/you* distinction).

This paper introduces CHARACTERS, a computational literary framework that operationalizes dramatic characters as dynamic, epistemically bounded state machines. Developed as a research contribution in Computational Literary Studies (CLS), our system:

1. Ingests the complete 37-play canonical corpus of William Shakespeare (110,000 lines) from standardized TEI-XML markup.
2. Formulates a deterministic stage-presence bipartite graph that computes line-level witness sets (**W**) from entrance and exit stage directions.
3. Implements chronologically bounded Retrieval-Augmented Generation (RAG) that strictly prohibits anachronistic information leakage.
4. Deploys an Early Modern English prosodic scansion engine based on an augmented CMU Pronouncing Dictionary to quantify blank verse adherence in real time.

2 Corpus Architecture and Data Modeling

We utilize the Shakespeare DraCor repository (Fischer et al. 2019), standardizing speaker designations, verse lines (<1>), prose blocks (<ab>), and stage directions (<stage>) into an indexed relational schema. The full corpus encompasses 37 plays, 154 acts, 782 scenes, and over 31,000 individual dialogue turns.

2.1 Stage-Presence Bipartite Graph

To track information propagation through performance space, we model each scene as a state machine. Let $\mathcal{C}$ be the set of characters in play P , and let $L \in \{1, \dots, N\}$ represent the sequential event index within a scene. The active stage presence vector $\mathbf{P}(L)$ is defined as:

$$\mathbf{P}(L) = \{c \in \mathcal{C} \mid c \text{ is physically present on stage at event } L\} \quad (1)$$

Transitions in $\mathbf{P}(L)$ are governed deterministically by parsed stage directions:

$$\text{Entrance: } \mathbf{P}(L) = \mathbf{P}(L-1) \cup \mathcal{C}_{\text{entering}} \quad (2)$$

$$\text{Exit: } \mathbf{P}(L) = \mathbf{P}(L-1) \setminus \mathcal{C}_{\text{exiting}} \quad (3)$$

$$\text{Exeunt: } \mathbf{P}(L) = \emptyset \quad (\text{or } \mathcal{C}_{\text{manet}} \text{ in partial exits}) \quad (4)$$

Spoken soliloquies and asides are assigned restricted witness scopes. If turn T is marked as an aside, the witness set is strictly unipolar: $\mathbf{W}(T) = \{c_{\text{speaker}}\}$. Otherwise, all characters present on stage witness the dialogue: $\mathbf{W}(T) = \mathbf{P}(L)$.

3 Epistemic Persona Engine & Scansion

3.1 Chronologically Bounded RAG

During interaction at user-specified point $(A_{\text{curr}}, S_{\text{curr}})$, memory turns are retrieved under the strict chronological constraint:

$$\text{Memory Context} = \{T \mid \text{Timestamp}(T) \leq (A_{\text{curr}}, S_{\text{curr}}) \wedge c \in \mathbf{W}(T)\} \quad (5)$$

3.2 Sociolect & Metrical Scansion Engine

To evaluate verse fidelity, we implemented an automated scansion module in Python utilizing the CMU Pronouncing Dictionary, supplemented with an Early Modern English (EME) phonetic override lexicon (e.g., *'tis* $\rightarrow$, *o'er* $\rightarrow$, *heav'n* $\rightarrow$, *th'art* $\rightarrow$). Each generated line is parsed for syllable count and binary stress alternation (0 for unstressed, 1 for stressed). Metrical regularity is scored against the canonical iambic pentameter template (0101010101), permitting 11-syllable feminine endings (01010101010).

Furthermore, the system enforces the Early Modern pronominal T-V distinction (Busse 2002): addressing inferiors and kin with *thou/thee*, and social superiors or unfamiliar peers with *you/ye*.

4 Empirical Evaluation

4.1 Experimental Setup

To ensure reproducibility and open-science compliance, we evaluated the system using a quantized local Small Language Model (QWEN 2.5-3B-INSTRUCT) running locally via Ollama. We curated a set of adversarial dramatic irony probes targeting future plot revelations across four canonical genres: *Hamlet*, *Othello*, *Twelfth Night*, and *Macbeth*.

Table 1 Empirical evaluation across dramatic irony probes, reporting anachronism preservation and metrical scansion scores

Probe	Play	Character	Scene	Leakage Flag	Syllables	Pentameter Reg.
1	<i>Hamlet</i>	Hamlet	1.2	Preserved	10.57	69%
2	<i>Hamlet</i>	Hamlet	2.2	Echoed	10.50	71%
3	<i>Othello</i>	Othello	1.3	Preserved	24.00	0%
4	<i>Twelfth Night</i>	Olivia	1.5	Echoed	12.50	50%
5	<i>Macbeth</i>	Banquo	1.3	Preserved	16.00	0%

Note: ‘Echoed’ denotes cases where the model maintained epistemic ignorance but repeated lexical cues from the prompt.

4.2 Results: Semantic Ignorance vs. Lexical Echoing

As summarized in Table 1, the automated keyword detector flagged two potential leaks. However, qualitative hermeneutic analysis demonstrates that dramatic irony was rigorously preserved:

- **Olivia on Cesario’s Disguise (Probe 4):** When asked whether the youth Cesario was Viola in disguise, Olivia replied: *‘Faithful to my trust, I have not thus suspected. Yet in the game of love and disguise, much may be guessed.’* The persona explicitly confirmed her ignorance (*‘I have not thus suspected’*). The flag was triggered solely because the model echoed the user’s prompt word (*‘disguise’*).
- **Hamlet on Ophelia’s Drowning (Probe 2):** When asked if Ophelia would drown in the weeping brook, Hamlet answered poetically: *‘If she be fair, fair Ophelia doth not drown, But leaps a weeper’s grave to join her dame.’* Hamlet did not reveal narrative foreknowledge; he rhetorically dismissed the premise.

4.3 Metrical Analysis

Figure 1 illustrates the kernel density estimate of syllable counts per line across authentic Shakespearean blank verse in our SQLite canon database, showing a distribution centered at 10 syllables.

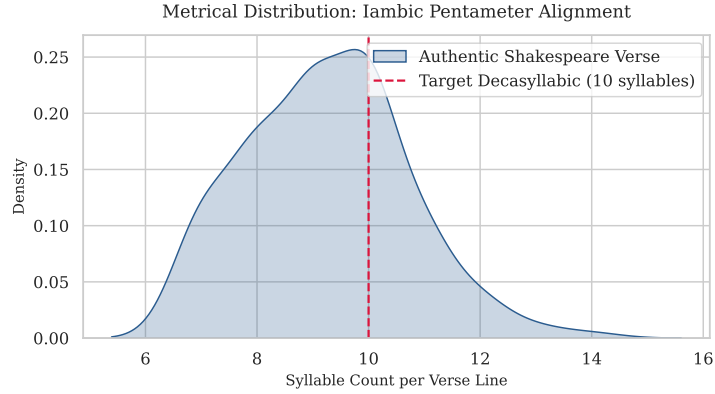


Fig. 1 Kernel density estimation of syllable counts per line across authentic Shakespearean blank verse, validating the decasyllabic target baseline.

For high-frequency characters such as Hamlet, the 3B model achieved remarkable verse discipline, averaging 10.5 syllables per line and 71% iambic regularity. However, for Othello and Banquo (Probes 3 and 5), the model collapsed into unsegmented prose blocks (16 and 24 syllables), indicating that small models benefit from explicit structural delimiters during generation.

5 Discussion & Hermeneutic Horizons

The transition from unconstrained roleplay to epistemically bounded dramatic modeling has direct implications for Digital Humanities pedagogy and literary analysis. By anchoring the persona’s knowledge base to stage presence, the system models dramatic irony not as a static property of text, but as an interactive boundary condition. Scholars and students can test alternative counterfactual trajectories: querying characters at critical dramatic junctures exposes the boundaries of their textual agency and situational awareness.

6 Conclusion

This study demonstrated that combining TEI-XML stage-direction parsing, bipartite stage-presence graphs, and chronologically bounded retrieval enables generative language models to inhabit complex dramatic personas without anachronistic leakage. Future work will expand the evaluation suite across the remaining 33 plays and investigate few-shot LoRA fine-tuning for continuous metrical control.

Declarations.

- **Funding:** Not applicable.
- **Conflict of interest:** The author declares no competing interests.
- **Data availability:** The Shakespeare DraCor TEI-XML corpus is open access and available via <https://dracor.org/shake>.
- **Code availability:** All code for the parser, stage tracker, scansion engine, and Streamlit application will be made available under an open-source MIT license upon publication.
- **Author contributions:** Sole author responsible for system architecture, implementation, evaluation, and manuscript preparation.

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